

standardization

June 6, 2024

```
[10]: import numpy as np # linear algebra
import pandas as pd # data processing
import matplotlib.pyplot as plt
import seaborn as sns
```

```
[11]: df = pd.read_csv('Social_Network_Ads.csv')
```

```
[12]: df=df.iloc[:,2:]
```

```
[13]: df.sample(5)
```

```
[13]:
```

	Age	EstimatedSalary	Purchased
32	21	16000	0
219	59	143000	1
166	22	55000	0
94	29	83000	0
360	43	129000	1

0.1 Train test split

```
[14]: from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(df.drop('Purchased',
↪axis=1),
                                                    df['Purchased'],
                                                    test_size=0.3,
                                                    random_state=0)

X_train.shape, X_test.shape
```

```
[14]: ((280, 2), (120, 2))
```

1 StandardScaler

```
[15]: from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()
```

```
# fit the scaler to the train set, it will learn the parameters
scaler.fit(X_train)

# transform train and test sets
X_train_scaled = scaler.transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

```
[16]: scaler.mean_
```

```
[16]: array([3.78642857e+01, 6.98071429e+04])
```

```
[17]: X_train
```

```
[17]:      Age  EstimatedSalary
92     26             15000
223    60            102000
234    38            112000
232    40            107000
377    42             53000
..     ..             ...
323    48             30000
192    29             43000
117    36             52000
47     27             54000
172    26            118000
```

```
[280 rows x 2 columns]
```

```
[18]: X_train_scaled
```

```
[18]: array([[ -1.1631724 , -1.5849703 ],
          [ 2.17018137,  0.93098672],
          [ 0.0133054 ,  1.22017719],
          [ 0.20938504,  1.07558195],
          [ 0.40546467, -0.48604654],
          [-0.28081405, -0.31253226],
          [ 0.99370357, -0.8330751 ],
          [ 0.99370357,  1.8563962 ],
          [ 0.0133054 ,  1.24909623],
          [-0.86905295,  2.26126285],
          [-1.1631724 , -1.5849703 ],
          [ 2.17018137, -0.80415605],
          [-1.35925203, -1.46929411],
          [ 0.40546467,  2.2901819 ],
          [ 0.79762394,  0.75747245],
          [-0.96709276, -0.31253226],
          [ 0.11134522,  0.75747245],
```

[-0.96709276, 0.55503912],
 [0.30742485, 0.06341534],
 [0.69958412, -1.26686079],
 [-0.47689368, -0.0233418],
 [-1.7514113 , 0.3526058],
 [-0.67297331, 0.12125343],
 [0.40546467, 0.29476771],
 [-0.28081405, 0.06341534],
 [-0.47689368, 2.2901819],
 [0.20938504, 0.03449629],
 [1.28782302, 2.20342476],
 [0.79762394, 0.26584866],
 [-0.28081405, 0.15017248],
 [0.0133054 , -0.54388463],
 [-0.18277423, 0.15017248],
 [-0.08473441, 0.23692961],
 [0.0133054 , -0.25469417],
 [2.17018137, 1.104501],
 [-1.7514113 , 0.3526058],
 [1.87606192, 0.12125343],
 [0.40546467, -0.13901799],
 [-1.1631724 , 0.29476771],
 [0.79762394, 1.36477242],
 [-0.28081405, -0.25469417],
 [-1.65337148, -0.05226085],
 [-0.96709276, -0.74631796],
 [0.30742485, 0.49720103],
 [-0.08473441, -1.06442747],
 [-1.06513258, 0.58395817],
 [0.11134522, -0.80415605],
 [-0.96709276, 1.53828669],
 [-0.67297331, 1.39369146],
 [-1.26121221, 0.49720103],
 [-0.28081405, 0.03449629],
 [-0.08473441, 0.00557724],
 [-0.28081405, -0.89091319],
 [0.89566375, -1.35361793],
 [-0.28081405, 2.2323438],
 [0.99370357, 1.97207239],
 [-1.1631724 , 0.46828198],
 [-1.26121221, 0.26584866],
 [1.38586284, 1.97207239],
 [1.28782302, -1.35361793],
 [-0.28081405, -0.28361322],
 [-0.47689368, 1.24909623],
 [-0.77101313, 1.07558195],
 [0.99370357, -1.06442747],

[0.30742485, 0.29476771],
 [0.99370357, 0.75747245],
 [-0.67297331, -1.49821316],
 [-0.67297331, 0.03449629],
 [0.50350449, 1.71180097],
 [2.07214155, 0.17909152],
 [-1.94749093, -0.74631796],
 [-0.18277423, 1.39369146],
 [0.40546467, 0.58395817],
 [0.89566375, -1.1511846],
 [-1.1631724 , -0.775237],
 [0.20938504, 0.23692961],
 [0.79762394, -0.31253226],
 [2.07214155, -0.80415605],
 [0.79762394, 0.12125343],
 [-0.28081405, 0.61287722],
 [-0.96709276, -0.31253226],
 [0.20938504, -0.37037036],
 [2.07214155, 2.11666762],
 [1.87606192, -1.26686079],
 [1.38586284, -0.91983223],
 [0.89566375, 1.24909623],
 [1.48390265, 2.11666762],
 [-0.28081405, -1.23794174],
 [1.97410174, 0.90206768],
 [0.69958412, -0.71739891],
 [-1.45729185, 0.3526058],
 [0.79762394, -1.35361793],
 [0.40546467, -0.13901799],
 [-0.96709276, 0.41044389],
 [0.0133054 , -0.31253226],
 [-1.1631724 , 0.41044389],
 [-0.86905295, -1.2090227],
 [-0.08473441, 0.03449629],
 [-1.55533166, -0.42820845],
 [0.99370357, -1.00658937],
 [1.09174339, -1.2090227],
 [0.0133054 , -0.13901799],
 [-1.06513258, -1.52713221],
 [0.79762394, -1.2090227],
 [0.99370357, 2.05882953],
 [-1.1631724 , -1.52713221],
 [-0.28081405, 0.78639149],
 [0.11134522, -0.31253226],
 [-1.35925203, -1.23794174],
 [-0.5749335 , -1.49821316],
 [0.79762394, 0.52612008],

[-0.28081405, -0.34145131],
 [1.7780221 , -0.28361322],
 [0.89566375, -1.03550842],
 [0.20938504, 0.06341534],
 [-0.5749335 , 0.87314863],
 [-1.84945111, -1.41145602],
 [-1.26121221, 0.58395817],
 [-0.28081405, 0.52612008],
 [-0.96709276, -1.09334651],
 [1.1897832 , -1.44037507],
 [0.20938504, -0.31253226],
 [1.1897832 , -0.74631796],
 [-0.28081405, 0.06341534],
 [0.20938504, 2.08774857],
 [0.79762394, -1.09334651],
 [0.11134522, 0.03449629],
 [-1.7514113 , 0.12125343],
 [-0.86905295, 0.15017248],
 [-0.67297331, 0.17909152],
 [0.89566375, -1.29577984],
 [0.20938504, -0.25469417],
 [-0.37885386, 1.22017719],
 [0.0133054 , 0.29476771],
 [0.40546467, 0.15017248],
 [0.89566375, -0.65956082],
 [0.11134522, 0.15017248],
 [-1.84945111, -1.29577984],
 [-0.08473441, 0.29476771],
 [-0.18277423, -0.28361322],
 [0.30742485, -0.51496559],
 [-0.18277423, 1.59612479],
 [0.99370357, -1.18010365],
 [-0.18277423, 1.62504383],
 [1.28782302, 1.8563962],
 [-1.06513258, -0.37037036],
 [0.0133054 , 0.03449629],
 [0.11134522, -0.25469417],
 [-1.55533166, -1.23794174],
 [-0.47689368, -0.28361322],
 [0.99370357, 0.12125343],
 [1.97410174, -1.35361793],
 [1.48390265, 0.06341534],
 [-0.5749335 , 1.36477242],
 [1.58194247, 0.00557724],
 [-0.77101313, 0.29476771],
 [1.97410174, 0.7285534],
 [-1.1631724 , -0.51496559],

[0.69958412, 0.26584866],
 [-1.35925203, -0.42820845],
 [0.20938504, 0.15017248],
 [-0.47689368, -1.2090227],
 [0.6015443 , 2.00099143],
 [-1.55533166, -1.49821316],
 [-0.47689368, -0.54388463],
 [0.50350449, 1.82747716],
 [-1.35925203, -1.09334651],
 [0.79762394, -1.38253697],
 [-0.28081405, -0.42820845],
 [1.58194247, 0.98882482],
 [0.99370357, 1.42261051],
 [-0.28081405, -0.48604654],
 [-0.08473441, 2.14558666],
 [-1.45729185, -0.11009894],
 [-0.08473441, 1.94315334],
 [-0.67297331, -0.34145131],
 [-0.47689368, -0.8330751],
 [0.69958412, -1.38253697],
 [-0.77101313, -1.5849703],
 [-1.84945111, -1.46929411],
 [1.09174339, 0.12125343],
 [0.11134522, 1.50936765],
 [-0.28081405, 0.09233438],
 [0.11134522, 0.03449629],
 [-1.35925203, -1.35361793],
 [0.30742485, 0.06341534],
 [-0.86905295, 0.38152485],
 [1.58194247, -1.26686079],
 [-0.28081405, -0.74631796],
 [-0.08473441, 0.15017248],
 [-0.86905295, -0.65956082],
 [-0.67297331, -0.05226085],
 [0.40546467, -0.45712749],
 [-0.77101313, 1.88531525],
 [1.38586284, 1.27801528],
 [1.1897832 , -0.97767033],
 [1.7780221 , 1.82747716],
 [-0.86905295, -0.25469417],
 [-0.77101313, 0.55503912],
 [-1.1631724 , -1.55605125],
 [-0.47689368, -1.12226556],
 [0.30742485, 0.06341534],
 [-0.18277423, -1.06442747],
 [1.67998229, 1.59612479],
 [0.99370357, 1.76963906],

[0.30742485, 0.03449629],
 [-0.77101313, -0.22577513],
 [-0.08473441, 0.06341534],
 [0.30742485, -0.19685608],
 [1.97410174, -0.65956082],
 [-0.77101313, 1.33585337],
 [-1.7514113 , -0.60172273],
 [-0.08473441, 0.12125343],
 [0.30742485, -0.31253226],
 [1.09174339, 0.55503912],
 [-0.96709276, 0.26584866],
 [1.48390265, 0.3526058],
 [0.20938504, -0.37037036],
 [2.17018137, -1.03550842],
 [-0.28081405, 1.104501],
 [-1.65337148, 0.06341534],
 [0.0133054 , 0.03449629],
 [0.11134522, 1.04666291],
 [-0.08473441, -0.37037036],
 [-1.1631724 , 0.06341534],
 [-0.28081405, -1.35361793],
 [1.58194247, 1.104501],
 [-0.77101313, -1.52713221],
 [0.11134522, 1.8563962],
 [-0.86905295, -0.775237],
 [-0.47689368, -0.775237],
 [-0.28081405, -0.91983223],
 [0.30742485, -0.71739891],
 [0.30742485, 0.06341534],
 [0.11134522, 1.8563962],
 [-1.06513258, 1.94315334],
 [-1.65337148, -1.55605125],
 [-1.1631724 , -1.09334651],
 [-0.67297331, -0.11009894],
 [0.11134522, 0.09233438],
 [0.30742485, 0.26584866],
 [0.89566375, -0.57280368],
 [0.30742485, -1.1511846],
 [-0.08473441, 0.67071531],
 [2.17018137, -0.68847986],
 [-1.26121221, -1.38253697],
 [-0.96709276, -0.94875128],
 [0.0133054 , -0.42820845],
 [-0.18277423, -0.45712749],
 [-1.7514113 , -0.97767033],
 [1.7780221 , 0.98882482],
 [0.20938504, -0.37037036],

```
[ 0.40546467,  1.104501  ],
[-1.7514113 , -1.35361793],
[ 0.20938504, -0.13901799],
[ 0.89566375, -1.44037507],
[-1.94749093,  0.46828198],
[-0.28081405,  0.26584866],
[ 1.87606192, -1.06442747],
[-0.37885386,  0.06341534],
[ 1.09174339, -0.89091319],
[-1.06513258, -1.12226556],
[-1.84945111,  0.00557724],
[ 0.11134522,  0.26584866],
[-1.1631724 ,  0.32368675],
[-1.26121221,  0.29476771],
[-0.96709276,  0.43936294],
[ 1.67998229, -0.89091319],
[ 1.1897832 ,  0.52612008],
[ 1.09174339,  0.52612008],
[ 1.38586284,  2.31910094],
[-0.28081405, -0.13901799],
[ 0.40546467, -0.45712749],
[-0.37885386, -0.775237  ],
[-0.08473441, -0.51496559],
[ 0.99370357, -1.1511846 ],
[-0.86905295, -0.775237  ],
[-0.18277423, -0.51496559],
[-1.06513258, -0.45712749],
[-1.1631724 ,  1.39369146]])
```

```
[19]: X_train_scaled = pd.DataFrame(X_train_scaled, columns=X_train.columns)
      X_test_scaled = pd.DataFrame(X_test_scaled, columns=X_test.columns)
```

```
[20]: np.round(X_train.describe(), 1)
```

```
[20]:
```

	Age	EstimatedSalary
count	280.0	280.0
mean	37.9	69807.1
std	10.2	34641.2
min	18.0	15000.0
25%	30.0	43000.0
50%	37.0	70500.0
75%	46.0	88000.0
max	60.0	150000.0

```
[21]: np.round(X_train_scaled.describe(), 1)
```



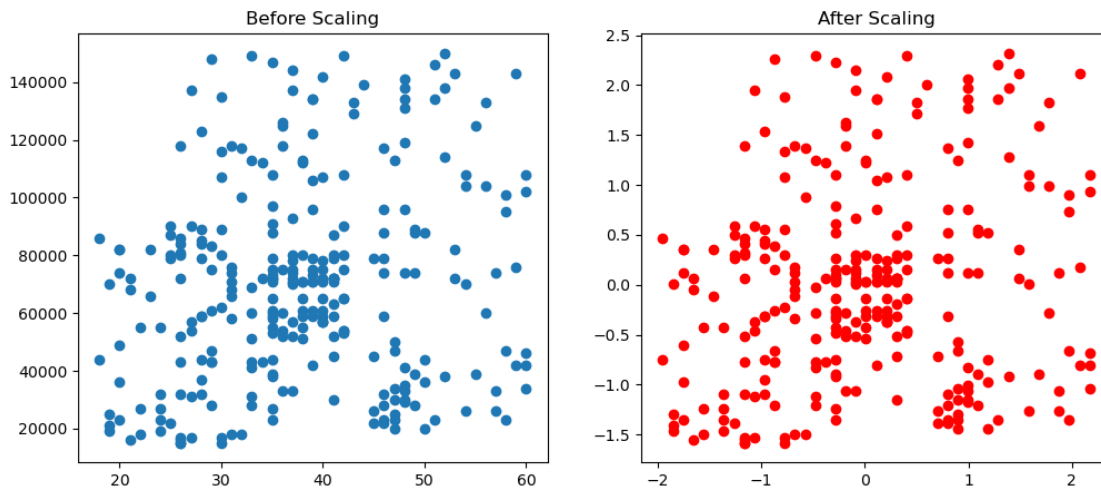
```
[21]:
```

	Age	EstimatedSalary
count	280.0	280.0
mean	0.0	0.0
std	1.0	1.0
min	-1.9	-1.6
25%	-0.8	-0.8
50%	-0.1	0.0
75%	0.8	0.5
max	2.2	2.3

2 Effect of Scaling

```
[22]: fig, (ax1, ax2) = plt.subplots(ncols=2, figsize=(12, 5))

ax1.scatter(X_train['Age'], X_train['EstimatedSalary'])
ax1.set_title("Before Scaling")
ax2.scatter(X_train_scaled['Age'], X_train_scaled['EstimatedSalary'], color='red')
ax2.set_title("After Scaling")
plt.show()
```

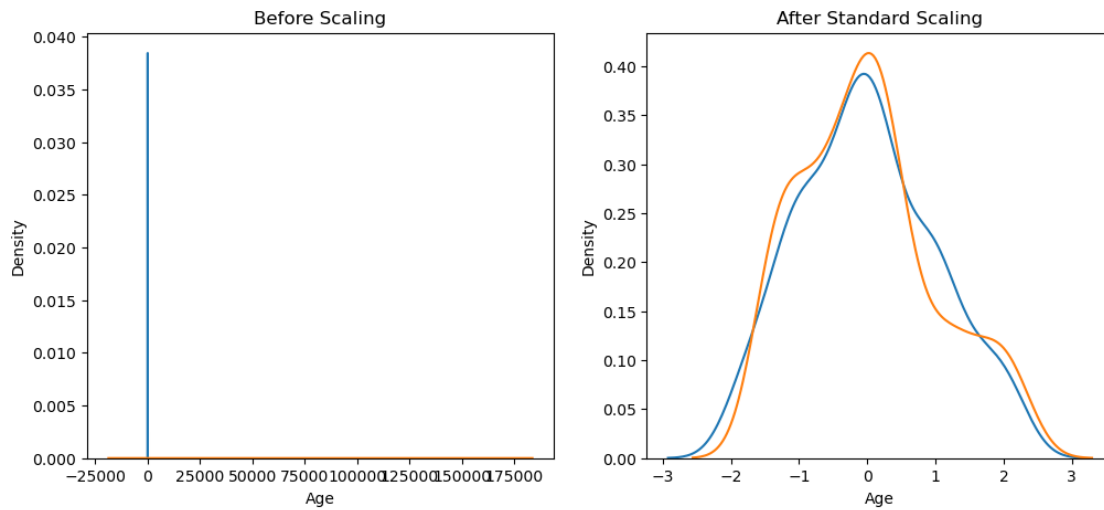


```
[23]: fig, (ax1, ax2) = plt.subplots(ncols=2, figsize=(12, 5))

# before scaling
ax1.set_title('Before Scaling')
sns.kdeplot(X_train['Age'], ax=ax1)
sns.kdeplot(X_train['EstimatedSalary'], ax=ax1)

# after scaling
```

```
ax2.set_title('After Standard Scaling')
sns.kdeplot(X_train_scaled['Age'], ax=ax2)
sns.kdeplot(X_train_scaled['EstimatedSalary'], ax=ax2)
plt.show()
```

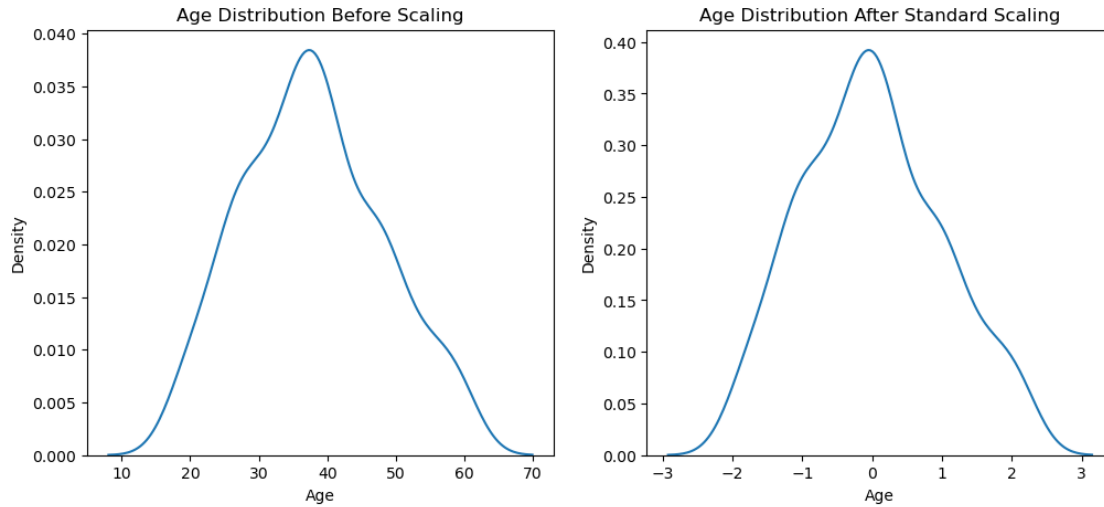


3 Comparison of Distributions

```
[24]: fig, (ax1, ax2) = plt.subplots(ncols=2, figsize=(12, 5))

# before scaling
ax1.set_title('Age Distribution Before Scaling')
sns.kdeplot(X_train['Age'], ax=ax1)

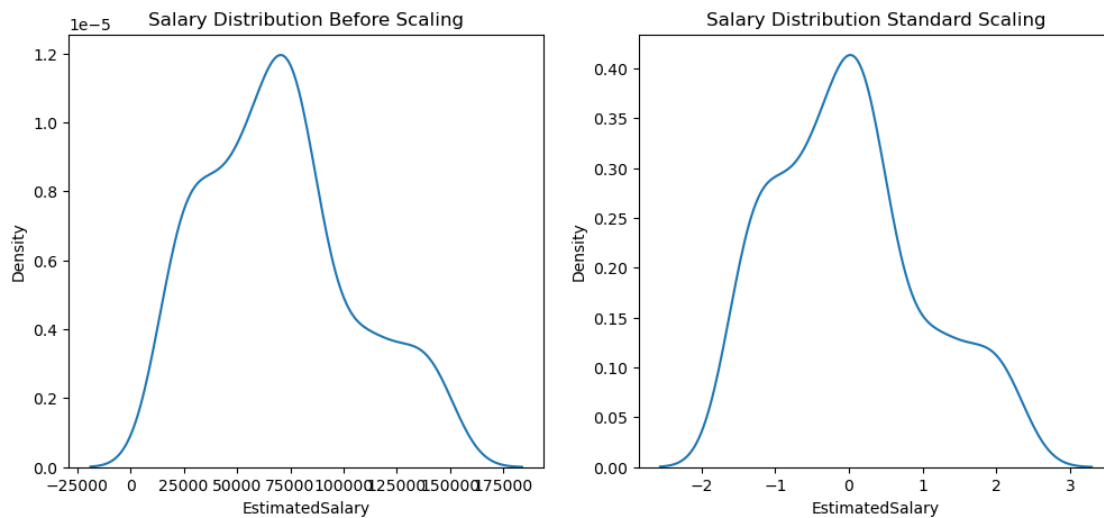
# after scaling
ax2.set_title('Age Distribution After Standard Scaling')
sns.kdeplot(X_train_scaled['Age'], ax=ax2)
plt.show()
```



```
[25]: fig, (ax1, ax2) = plt.subplots(ncols=2, figsize=(12, 5))
```

```
# before scaling
ax1.set_title('Salary Distribution Before Scaling')
sns.kdeplot(X_train['EstimatedSalary'], ax=ax1)

# after scaling
ax2.set_title('Salary Distribution Standard Scaling')
sns.kdeplot(X_train_scaled['EstimatedSalary'], ax=ax2)
plt.show()
```



```
[ ]:
```